Figure 4.29 and 4.30 shows the picture of the modern electric water pump. It consumes less power and run smoothly without noise. This type of pump works to a level of water below 1000 feet. These types of pumps are commonly used in all places.

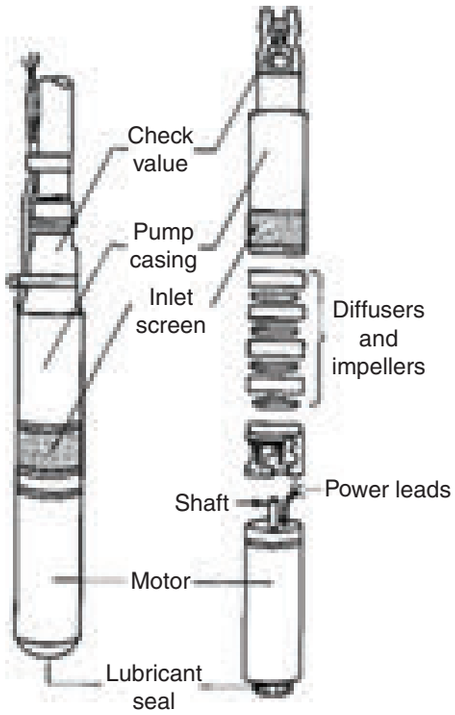


Fig. 4.30 Inner system

i. Selecting methods

Jet pumps are used to suck water to a depth of 500 feet in deep wells. According to the quantity of water required, the motor should be selected. Proper selection of the motor consumes less electric power and time.

In the present modern technology, 1 HP motor is commonly used. This will function up to 200 feet, and 100 liters of water can be obtained per minute. If we use 1.5 HP motor, it can deliver water to a level up to 500 feet in bore wells. If the motor is placed below 350 feet, only 45 liters of water will be delivered. But the cost of the electric motor is double.

ii. Installation of water pump

The good quality of motor and PVC pipes has to be selected and fit it in required depth. The air valve should be placed correctly. Otherwise, the electric motor is rotated in reverse. It will damage the bearing and more maintenance is required for the motor.

iii. Use of water pump

This type of pump is used in places where the water level is below 1000 feet.

4.4.3 Air compressor

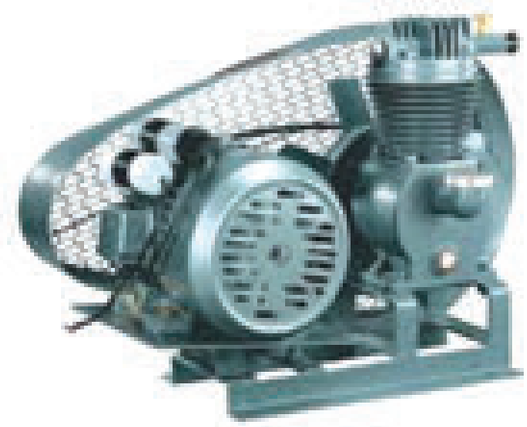


Fig. 4.31 Air compressor

The air compressor shown in figure 4.31 is a water pump used in the bore well. It creates water bubbles when going into the foot valve with air pressure. It has slightly special features than jet pump. It is used to pump water up to 300 feet with a capacity of 2 HP. The

BEE_ENG_Chapter_04.indd 68 12/27/2019 10:27:25 AM